Irradiance vs Wind Speed

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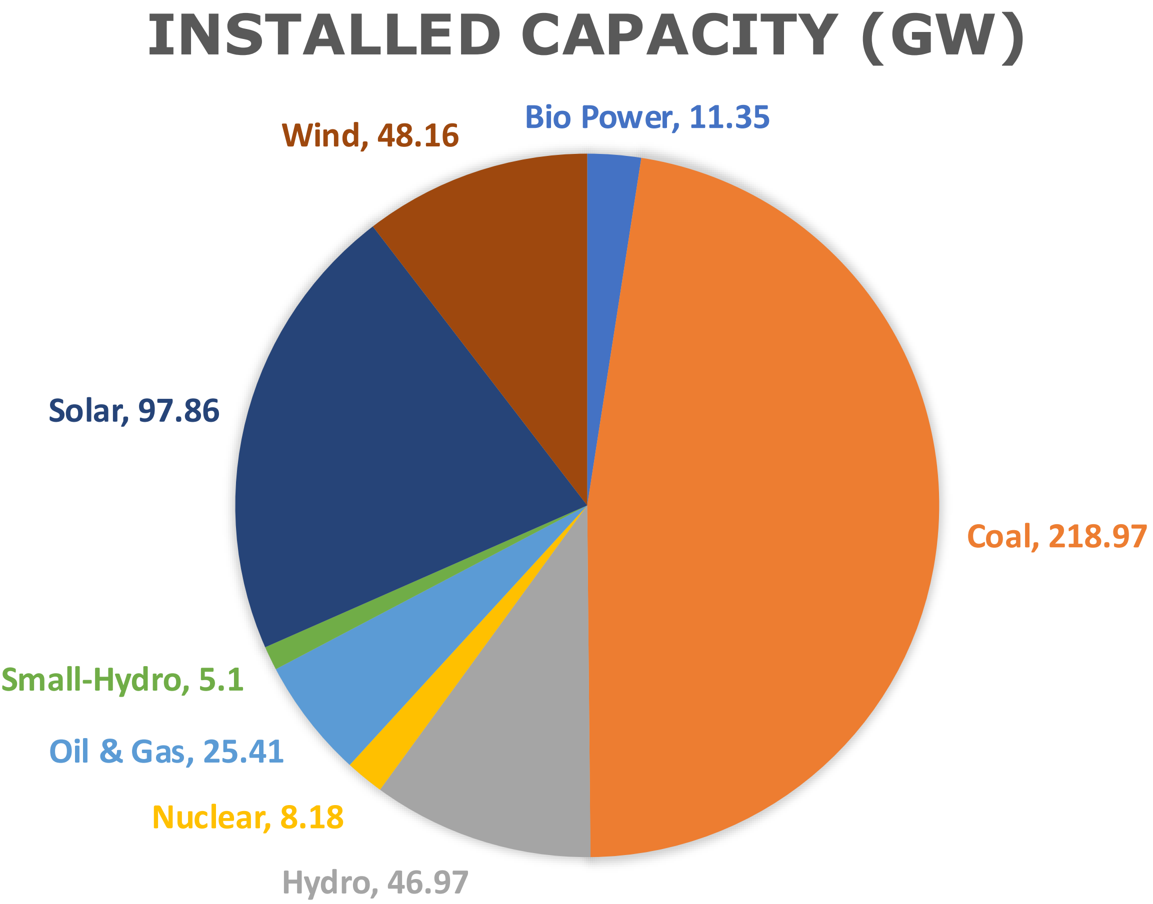
Why,

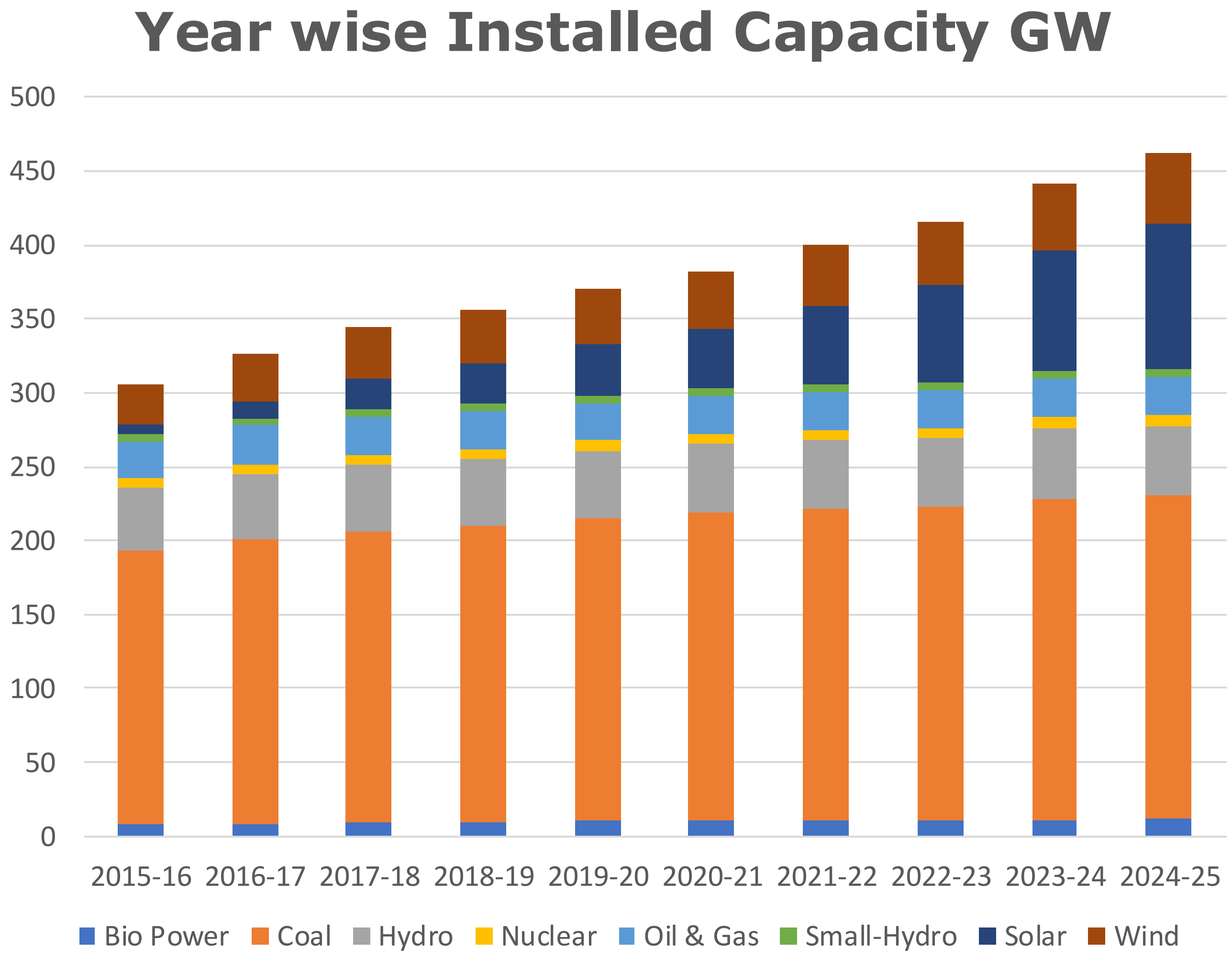
I tried to figure out who is really making money in the power industry, where substations involve far more technology than just solar panels and wind turbines.

However, as I delved deeper, I realized that the real question isn't about who is making money now, but rather who will dominate the industry in the future.

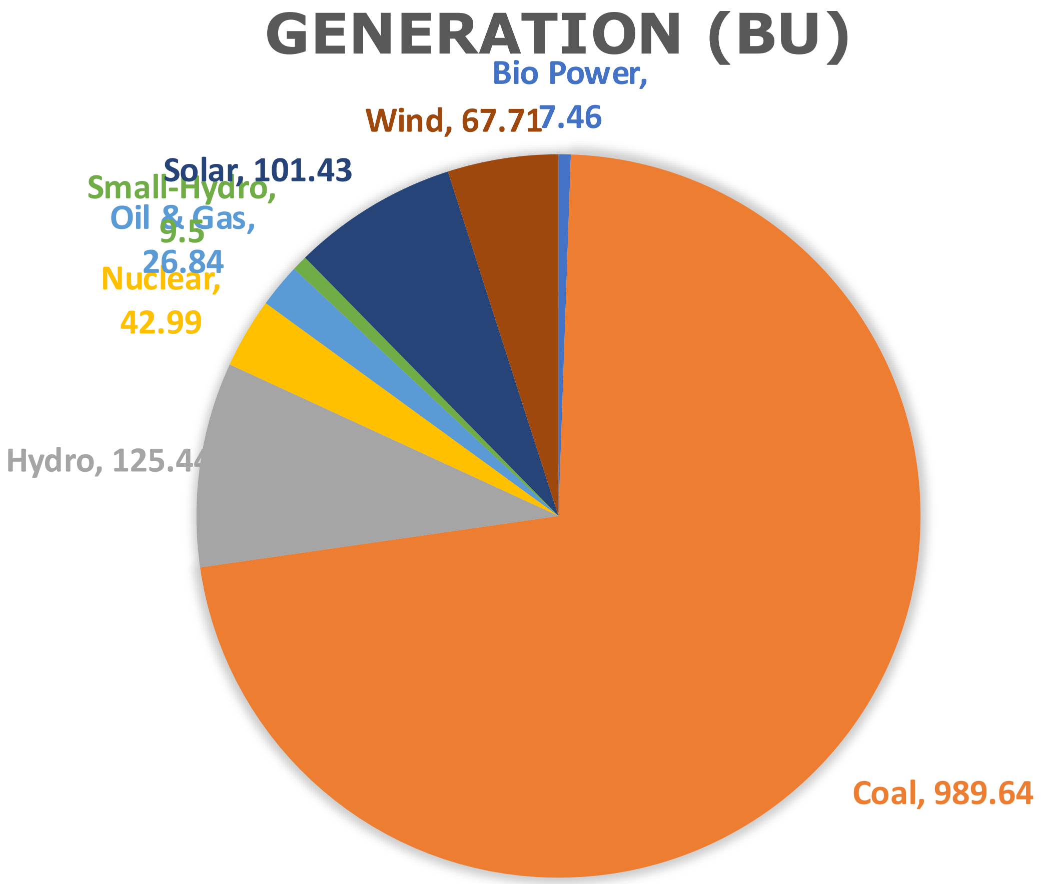
To explore this, I have put together this small presentation.

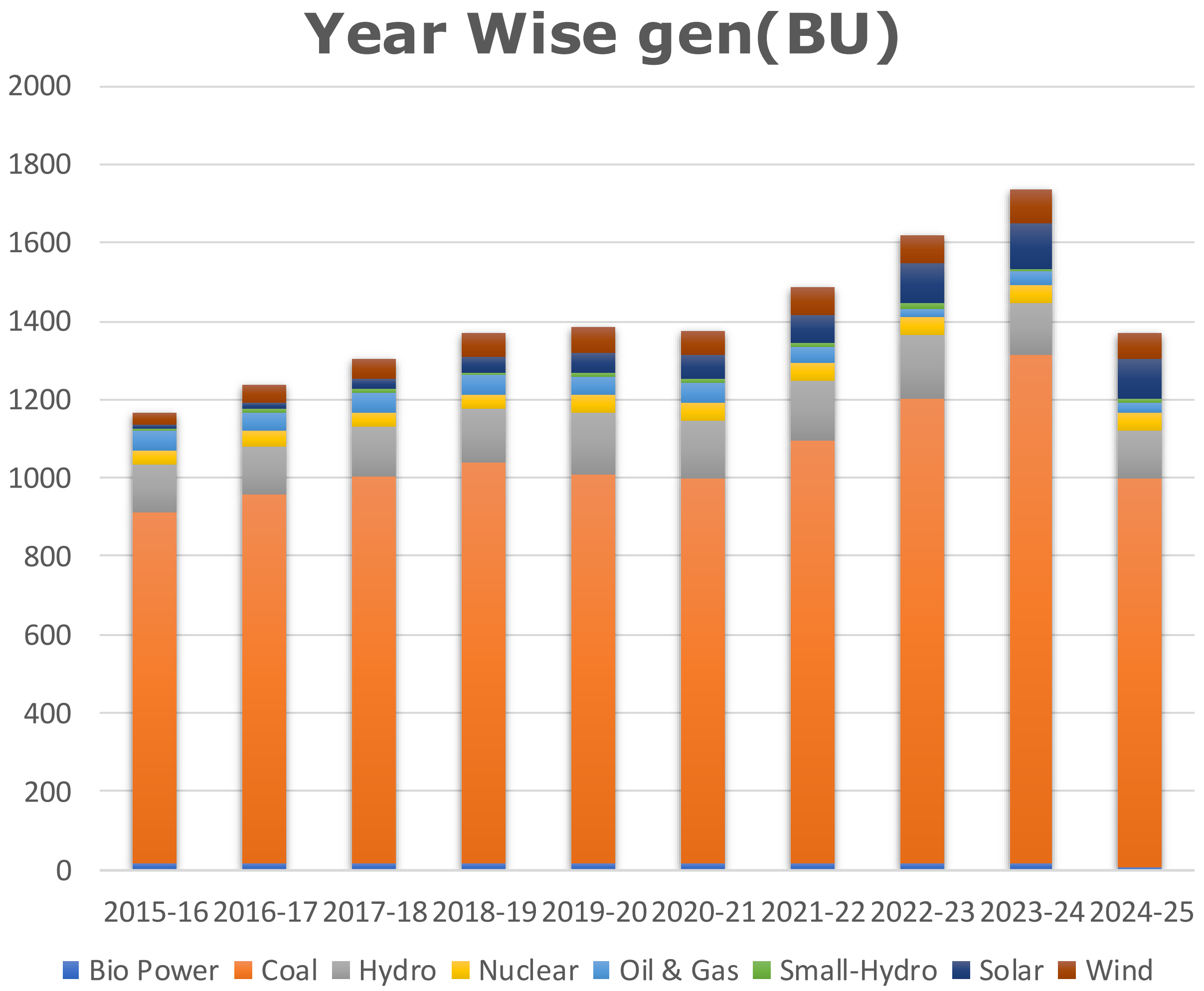
Power Installed Capacity





Power Generation Capacity



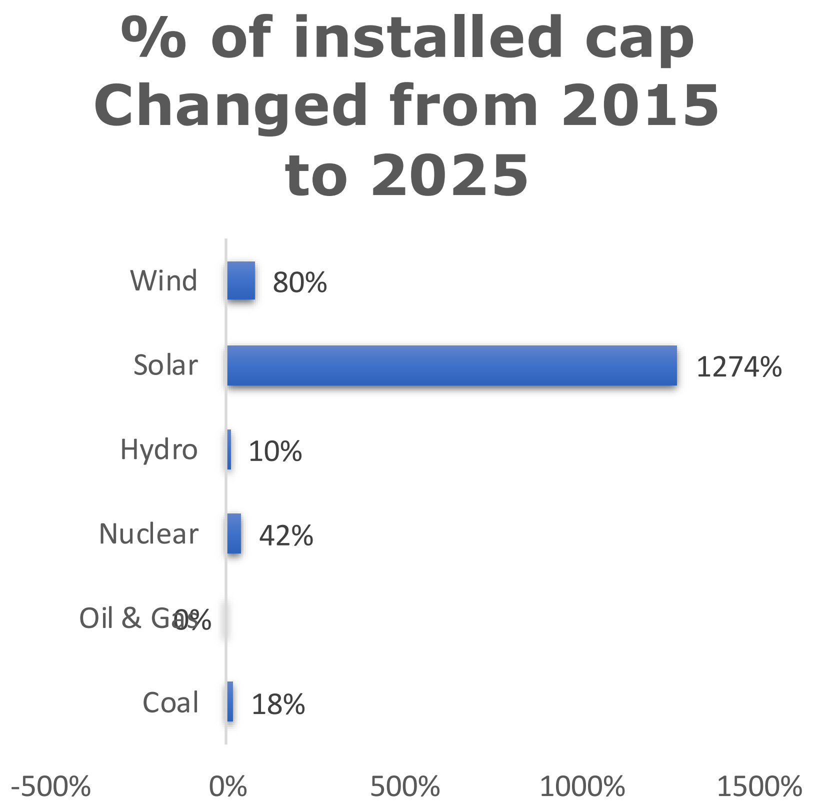


1 BU = 1 TWh = 10910^9109 kWh (Kilowatt-hours) or 1 BU ≈ 114.2 MW

Power Installed Capacity Outcomes

Capacity in 2016 vs 2025

Parameter	2015-16	2024-25	Change in GW	% Changed from 2015 to 2025
Coal (in GW)	185.17	218.97	33.80	18%
Oil & Gas (in GW)	25.50	25.41	-0.09	0%
Nuclear (in GW)	5.78	8.18	2.40	42%
Hydro (in GW)	42.78	46.97	4.18	10%
Solar (in GW)	7.12	97.86	90.74	1274%
Wind (in GW)	26.78	48.16	21.39	80%

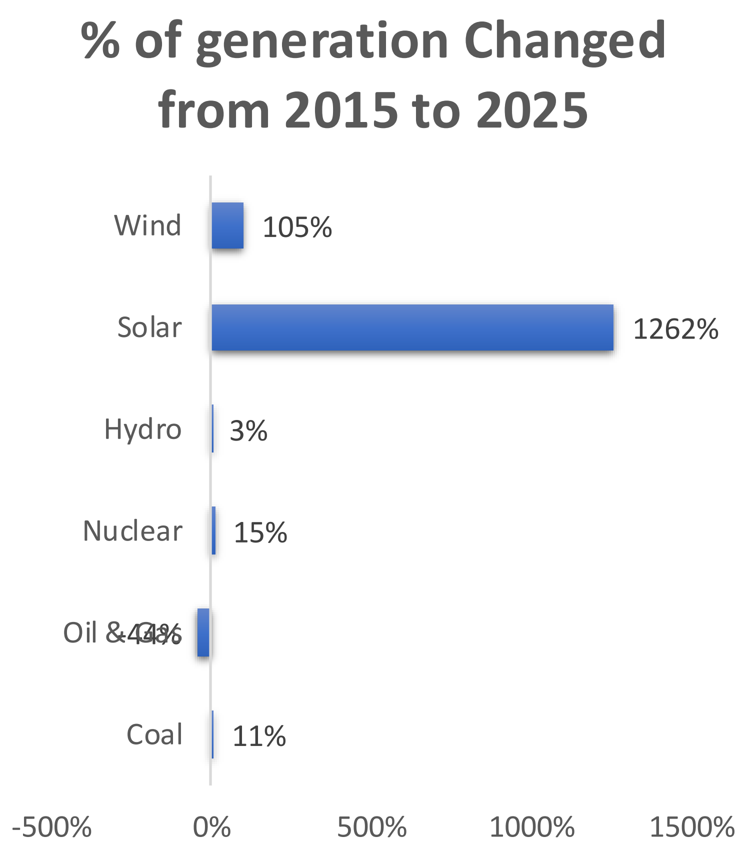


In the last 10 years, the Solar installed capacity has increased more than 10 times(90.74GW). However, coal also added its capacity up to 33GW.

Power Generation Capacity Outcomes

Generation in 2016 vs 2025

Parameter	2015-16	2024-25	Change in BU	% Changed from 2015 to 2025
Coal - Generation (In BU)	895.340	989.644	94.304	11%
Oil & Gas - Generation (In BU)	47.673	26.840	-20.833	-44%
Nuclear - Generation (In BU)	37.414	42.990	5.577	15%
Hydro - Generation (In BU)	121.377	125.443	4.066	3%
Solar - Generation (In BU)	7.448	101.430	93.982	1262%
Wind - Generation (In BU)	33.029	67.714	34.685	105%



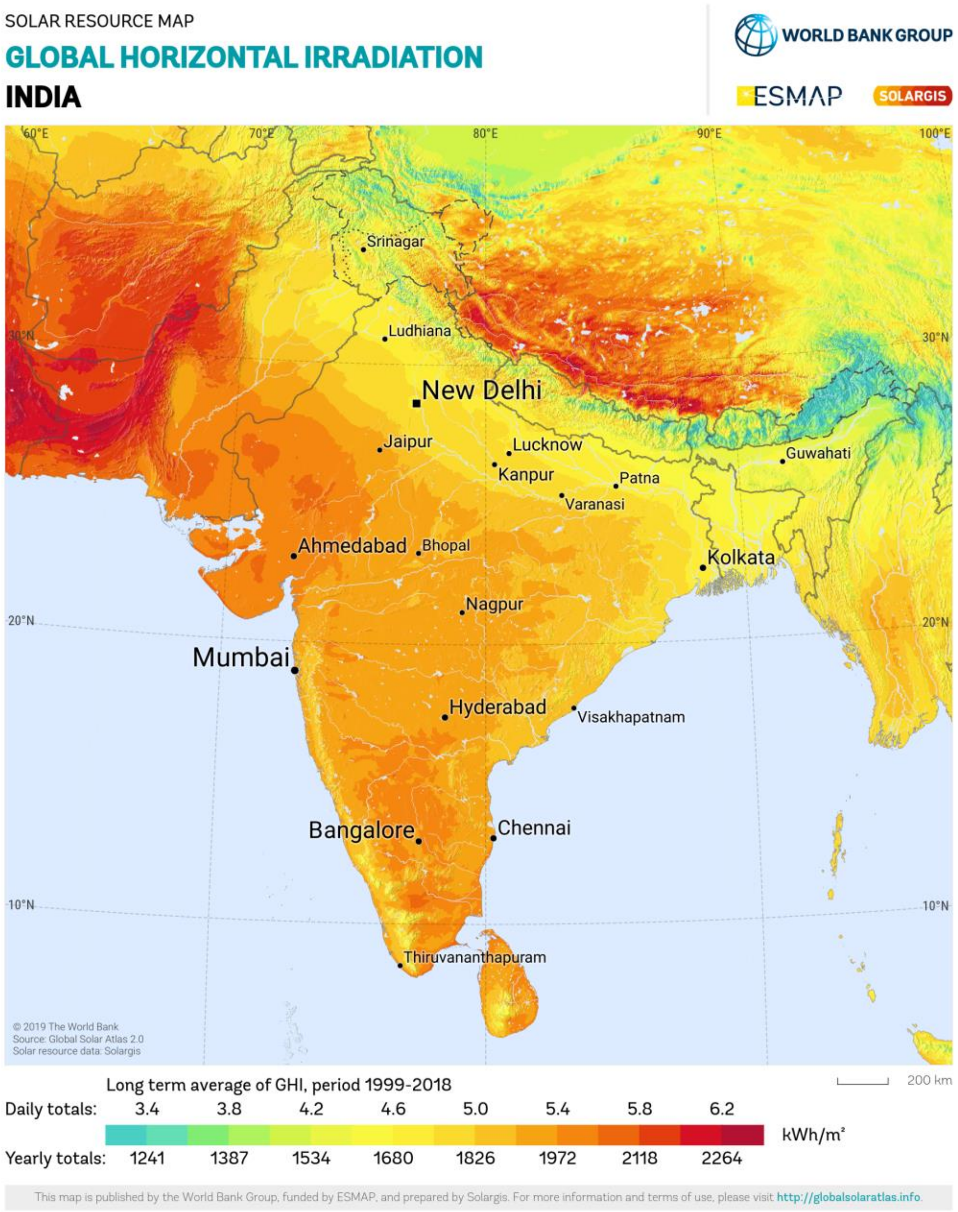
Here Solar change generation of power stood at 94BU and the same Coal also increased its generation by more than 94BU.

The Formula Trusted by Industry Experts

Solar + Wind + Storage = Future Power

- Nuclear is a strong renewable source, but the current focus is on solar, wind, and batteries.
- Hence, The best investment opportunities are in solar, wind, and battery storage.
- Solar capacity increased from 7 GW in 2015 to 98 GW in 2025, showing significant growth.
- Wind capacity grew from 26 GW in 2015 to 48 GW in 2025, with a slower expansion.
- Storage technology is still in the development stage, but solar and wind are leading the transition.

Potential for Wind and Solar



India's Solar Energy Potential

India receives **300 sunny days per year**, making it an ideal location for solar energy generation. According to the **National Institute of Solar Energy**, the country has the potential to generate **750 GW of solar power**, which is more than enough to meet its energy needs.

As of Jan 2025, India's solar power generation capacity is approximately 100.33 GW.
Ground-mounted solar plants: 76.95 GW
Grid-connected rooftop solar: 16.28 GW
Hybrid projects (solar component): 2.85 GW
Off-grid solar: 4.25 GW

Note: Still China holds 80% of global solar panel production.

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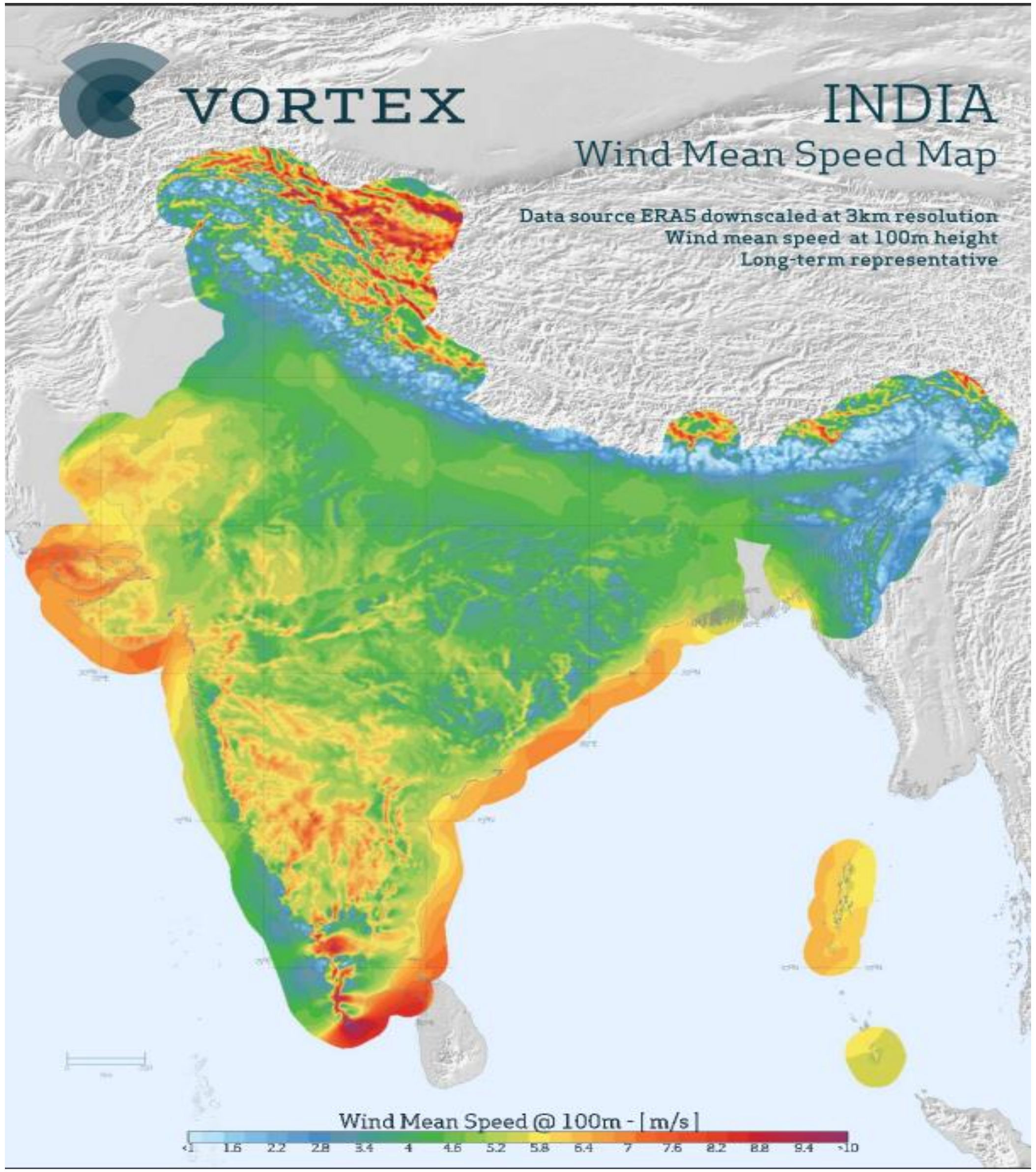

India’s Wind Energy Potential

The total installable wind potential in India is estimated to be **1164 GW at 150m AGL (Above Ground Level)**. This potential is distributed across different land types, with **544 GW available in wasteland, 607 GW in cultivable land, and 12 GW in forest land.**

As of November 2024, India's cumulative wind power generation capacity is approximately 47.96 GW.

Hence still Wind has 24.25x of Potential. Where solar has 8x.

Note: In the above, I think they didn’t include off-shore wind. Well, we can add 195GW more if we include this too.

Solar vs Wind

What happens in we install 1MW of wind and solar

SOLAR	WIND
A 1MW solar power plant can produce approximately 1.4 million kWh of electricity annually.	A 1MW wind turbine can produce approximately 3 million kWh of electricity annually.
The cost of a 1 MW solar power plant in India is estimated to be between INR 4.5 crores and INR 6 crores.	The cost of a 1 MW wind power plant in India can range from about Rs 5–6 crores.
The annual maintenance cost for a solar plant typically ranges between ₹8 lakhs to ₹12 lakhs per megawatt (MW).	The annual maintenance cost for a wind turbine in India is typically around ₹ 10 lakhs per MW.
The average lifespan of a solar panel is 25 to 30 years.	The expected lifespan of a wind turbine is around 20–30 years.
India imported around $7 billion worth of solar equipment in the 2023-24 fiscal year	In FY 2023/24, India's wind turbine imports totaled only around 11.11 million USD compared to export volumes of 205 million USD

Conclusion

- Though nuclear power has a bright future, the government is focusing more on wind and solar to bridge the power consumption gap.
- Unlike solar, wind energy has seen limited capacity additions. However, while increased solar panel purchases indirectly benefit China, India remains a net exporter of wind energy.
- For an idea, China has a 1,000 GW solar panel manufacturing capacity, whereas India has only 50-60 GW.
- For perspective, Chinese solar panels cost ₹12-14/Watt, whereas Indian panels are priced at ₹21-23/Watt.
- With most companies racing toward solar, wind energy gives me greater confidence in terms of investment potential.
- Considering all these factors, the "Make in India" initiative will significantly boost the wind energy sector.